

Savitribai Phule Pune University, Pune Third Year Information Technology (2019 Course) 314451: Computer Network and Security		
Teaching Scheme:	Credit Scheme:	Examination Scheme:
Theory (TH) : 3 hrs/week	03 Credit	Mid_Semester : 30 Marks End_Semester : 70 Marks
Prerequisite Courses: 1. Basics of Computer Network		
Companion Course: 1. Cyber Security		
Course Objectives: To familiarize students with- 1. The application layer services, responsibilities and protocol. 2. Fathom wireless network and different wireless standards 3. Differences in different wireless networks and to learn different mechanism used at layers of wireless network. 4. The concept of network security. 5. Basic cryptographic techniques in application development. 6. Cyber security vulnerabilities & study typical threats to modern digital systems.		
Course Outcomes: On completion of the course, students will be able to– CO1: Know Responsibilities, services offered and protocol used at application layer of network CO2: Understand wireless network and different wireless standards. CO3: Recognize the Adhoc Network's MAC layer, routing protocol and Sensor network architecture. CO4: Define the principal concepts of network security and Understand network security threats, security services, and countermeasures CO5: Apply basic cryptographic techniques in application development. CO6: Gain a good comprehension of the landscape of cyber security Vulnerabilities & describe typical threats to modern digital systems.		
COURSE CONTENTS		
Unit I	APPLICATION LAYER	(06 hrs)
Client Server Paradigm: Communication using TCP and UDP, Peer to Peer Paradigm, Application Layer Protocols: DNS, FTP, TFTP, HTTP, SMTP, POP, IMAP, MIME, DHCP, TELNET.		

Mapping of Course Outcomes for Unit I	CO1	
Unit II	WIRELESS STANDARDS	(06 hrs)
Wireless LANs: Fundamentals of WLAN, Design goals, Characteristics, Network Architecture, IEEE 802.11: components in IEEE 802.11 network, Physical Layer, MAC Sub Layers : DCF, PCF, Hidden and exposed station problem, Frame format, Addressing Mechanism, IEEE 802.15.1 Bluetooth: Architecture, Layers, operational states, IEEE 802.16 WiMax: Services, Architecture, Layers, comparison between Bluetooth, IEEE 802.11 and IEEE 802.16.		
Mapping of Course Outcomes for Unit II	CO2	
Unit III	ADHOC AND WSN	(06 hrs)
Infrastructure Network and Infrastructure-less Wireless Networks, Issues in Adhoc Wireless Network, Adhoc Network MAC Layer: Design Issues, Design Goal, Classification, MACAW, Adhoc Network Routing Layer: Issues in Designing a Routing Protocol for Ad-hoc Wireless Networks – Classifications of Routing Protocols, DSDV, AODV, DSR, Applications of Sensor Network, Comparison with Ad Hoc Wireless Network, Sensor node architecture Issues and Challenges in Designing a Sensor Network, Classification of sensor network protocols, SENSOR NETWORK ARCHITECTURE: Layered Architecture, Clustered Architecture,		
Mapping of Course Outcomes for Unit III	CO3	
Unit IV	INTRODUCTION TO NETWORK SECURITY	(06 hrs)
Importance and Need for Security, Network Attacks- Passive, Active Network Security Threats: Unauthorized access, Distributed Denial of Service (DDoS) attacks, Man in the middle attacks, Concept of Security Principles: Confidentiality and Privacy , Authentication, Authorization and Access Control, Integrity, Non- repudiation, Stream Ciphers: Substitution Cipher – Mono alphabetic Cipher, Polyalphabetic Substitution Cipher., Transposition Cipher: Rail-Fence Block Ciphers modes: Electronic Code Book (ECB) Mode., Cipher Block Chaining (CBC) Mode., Cipher Feedback Mode (CFB) , Output Feedback (OFB) Mode.		
Mapping of Course Outcomes for Unit IV	CO4	
Unit V	CRYPTOGRAPHIC ALGORITHM	(06 hrs)
Mathematical preliminaries: Groups, Rings, Fields, Prime numbers, Symmetric key algorithms: Data Encryption Standards, Advanced Encryption Standard, Public Key Encryption and Hash function: RSA, Digital signatures, Digital Certificates and Public Key Infrastructure: Private Key Management, Diffie-Hellman key exchange, The PKIX Model		

Mapping of Course Outcomes for Unit V	CO5	
Unit VI	INTRODUCTION TO CYBER SECURITY	(06 hrs)
Introduction to Cyber Security: Basic Cyber Security Concepts, Layers of security, Vulnerability, Threat, Harmful Acts-Malware, Phishing, MIM Attack, DOS Attack, SQL Injection, Internet Governance – Challenges and Constraints, Computer Criminals, Assets and Threat, Motive of Attackers, Software attacks, hardware attacks, Cyber Threats-Cyber Warfare, Cyber Crime, Cyber Stalking, Cyber Terrorism, Cyber Espionage, Comprehensive Cyber Security Policy		
Mapping of Course Outcomes for Unit VI	CO6	
Text Books:		
1. Behrouz A. Forouzan, TCP/IP Protocol Suite, McGraw Hill Education, ISBN: 978-0-07-070652-1, 4th Edition. 2. C. Siva Ram Murthy, B. S. Manoj, Adhoc Wireless Networks: Architecture and Protocols, Pearson Education, ISBN: 978-81-317-0688-6, 1st Edition. 3. Atul Kahate Cryptography and Network Security, 3e, McGraw Hill Education, 4. B. A. Forouzan Cryptography and Network Security McGraw Hill Education 5. William Stallings Cryptography and Network Security: Principles and Practice, 4th Edition. 6. Nina Godbole and Sunit Belpure, Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Wiley		
Reference Books:		
1. Kazem Sohraby, Daniel Minoli, Taieb Znati, Wireless Sensor Networks: Technology, Protocols and Applications, Wiley India, ISBN: 9788126527304 2. Schneir, Bruce, “Applied Cryptography: Protocols and Algorithms” 3. Charles E. Perkins, Adhoc Networking, Pearson Education, 978-81-317-2096-7 4. Andrew S. Tanenbaum, David J. Wethrall, Computer Network, Pearson Education, ISBN: 978-0-13-212695-3. 5. Kurose Ross, Computer Networking: A Top Down Approach Featuring the Internet, Pearson Education, ISBN: 978-81-7758-878- 6. Dr. V.K. Pachghare, Cryptography and Information security, PHI, Second edition, ISBN- 978-81-203-5082-3		
E- Books / E- Learning References :		
1. https://nptel.ac.in/courses/106/105/106105160/ 2. https://nptel.ac.in/courses/106/105/106105031/ 3. An Introduction to Cyber Security A Beginner’s Guide		